

Paper X — The Lepton Sector: Higgs-Free Hypercharge, the Koide Equipartition as an Electromagnetically-Selected 45° Critical Point, and the See-Saw Neutrino

B. Ludwig *Independent researcher*

Series: “A Universe in a Bottle” — Paper X of XI. Specialises the fermion sector (Paper IX) to the colourless leptons; builds on Papers III (mass / Koide), VI (weak isospin), and the Majorana extension of the weak sector.

New for the eleven-paper series (2026-06-08): the ten-paper draft folded leptons into the fermion paper; this paper gives the charged-lepton and neutrino sectors a dedicated, full treatment.

Abstract

We assemble the lepton sector of the BCC quantum-cellular-automaton model: the colourless, $SU(2)_L$ -charged fermions whose masses arise — like all masses in the model — from the chiral mass step of Paper III, with no Higgs field. Three results organise the sector. First, hypercharge is carried Higgs-free: the right-handed singlets are dynamical $U(1)_Y$ fields whose hypercharge difference $\Delta Y = Y_L - Y_R$ is absorbed onto the same gauge field $U(\mathbf{x})$ that carries the mass phase, so no scalar is needed and a pure ν_L state carries $T_3 = +\frac{1}{2}$ exactly. Second, the charged-lepton masses obey the Koide relation $Q = \sum m / (\sum \sqrt{m})^2 = 2/3$ to 0.91σ ($Q = 0.6666605$), which we show is exactly a 45° equipartition of the cubic amplitude $\sqrt{m} = y$ via the closed map $Q(\phi) = 1/(3 \cos^2 \phi)$; the use of $\sqrt{m}$ is the Cooper-pair bilinear $m = y^2$, the 45° is the *same* $SO(2)$ equipartition that caps the bound pair, and electromagnetism is the *selector* that puts only the colour-free, charged leptons on the critical point (quarks are pushed off by QCD; neutrinos are neutral and unpinned). Third, the neutrino acquires a small mass through a Higgs-free Majorana mass step on ν_R : the QCA Majorana step is exactly R-unitary, lepton-number-violating, gauge-allowed only for $Y_{\nu_R} = 0$, and combined with the chiral Dirac mass reproduces the see-saw $m_\nu \approx M_D^2/M_R$ to machine precision across $M_R/M_D \in [3, 10^5]$ — explaining the smallness of the neutrino mass with a single large scale rather than an unnaturally tiny coupling. All structural results are verified to machine precision; the one honest residual (the *magnitude* of the 45° rotation) is recorded.

1. Introduction

Leptons are the colourless half of the matter content: the charged leptons e, μ, τ (electric charge -1 , weak isospin doublet partners of the neutrinos) and the neutrinos ν_e, ν_μ, ν_τ (neutral). In the Standard Model their masses come from Yukawa couplings to the Higgs, and the smallness of the neutrino mass is either an unexplained tiny Yukawa or a see-saw involving a heavy right-handed Majorana scale. The lepton

sector is also where the most precise unexplained mass relation in particle physics lives: the Koide relation $Q = \frac{2}{3}$, which the charged leptons satisfy to one part in 10^5 but which has no SM explanation.

This paper treats all three pieces on the model's Higgs-free footing. Section 2 fixes the lepton field content and the Higgs-free hypercharge. Section 3 derives the Koide relation as a cubic-vector equipartition and identifies its 45° amplitude with the bound-pair stability cap. Section 4 builds the Higgs-free Majorana/see-saw neutrino. Section 5 states the electromagnetic selection rule that explains why *only* the charged leptons equipartition. The lepton sector inherits the three-generation T_{1u} structure and the spherical mass shell from Paper IX, and the chiral $SU(2)_L$ from Paper VI.

2. The lepton doublet and Higgs-free hypercharge

The first-generation leptons are the left-handed doublet $L = (\nu_e, e)_L$ with $Y_L = -1$, the right-handed charged singlet e_R with $Y_{e_R} = -2$, and the right-handed neutral singlet ν_R with $Y_{\nu_R} = 0$ (Paper IX, §5). Electric charge follows from Gell-Mann-Nishijima $Q = T_3 + Y/2$, exact per state.

In the SM the right-handed singlets get their hypercharge coupling to the photon/ Z through the Higgs Yukawa structure. Here there is no Higgs, so the hypercharge must be carried directly. Finding F41 shows this is consistent: the chiral mass step that couples $\eta_L \leftrightarrow \chi_R$ changes hypercharge by $\Delta Y = Y_L - Y_R$, and that difference is absorbed onto the *same* $SU(2)$ gauge field $U(\mathbf{x})$ that carries the complex-mass phase. The right-handed singlets are then promoted to genuinely dynamical $U(1)_Y$ -coupled fields by a Stueckelberg-wrapped kinetic half-step (Finding F42), whose $U(1)_Y$ Ward identity holds to 1.8×10^{-15} . The net effect: the full lepton hypercharge structure is reproduced with no scalar, and a pure ν_L state carries $T_3 = +\frac{1}{2}$ to 1.1×10^{-16} . The would-be Higgs vacuum direction is the pure-gauge field $U(\mathbf{x})$ of Paper III, not a physical boson.

3. The charged-lepton masses: the Koide relation as a 45° equipartition

3.1 Koide as cubic-vector equipartition

The three charged leptons fill the cubic vector irrep T_{1u} of O_h (Paper IX, §3). Form the amplitude vector $\mathbf{y} = (\sqrt{m_e}, \sqrt{m_\mu}, \sqrt{m_\tau})$ and decompose it into its democratic (A_{1g} , along $\hat{n} = (1, 1, 1)/\sqrt{3}$) and traceless (T_{1u}) parts. Writing ϕ for the angle between $\mathbf{y}$ and $\hat{n}$, the perpendicular part contributes nothing to the signed sum, so $\sum y = \sqrt{3} |\mathbf{y}| \cos \phi$ and $\sum y^2 = |\mathbf{y}|^2$, giving the **exact closed map** (Finding F80)

$$Q(\phi) \equiv \frac{\sum y^2}{(\sum y)^2} = \frac{\sum m}{(\sum \sqrt{m})^2} = \frac{1}{3 \cos^2 \phi}$$

with three landmark angles:

ϕ	Q	meaning
0°	$1/3$	democratic floor (degenerate generations)
45°	$2/3$	equipartition — the charged-lepton point
$54.7356^\circ = \arccos \frac{1}{\sqrt{3}}$	1	single-axis ceiling (one massive generation)

The measured charged leptons give $Q = 0.6666605$ (0.91σ from $\frac{2}{3}$), $\phi_{\text{lepton}} = 44.99974^\circ$, $|A_{1g}|^2/|T_{1u}|^2 = 1.00002$, coefficient of variation $\text{CV}(y) = 0.99999$. So $Q = \frac{2}{3}$ is *exactly* a 45° rotation of the mass amplitude away from democracy, equivalently the equal-weight condition $|A_{1g}| = |T_{1u}|$. It is also the exact midpoint of the allowed range, $Q = \frac{2}{3} = \frac{1}{2}(\frac{1}{3} + 1)$ — the maximally balanced point between “all generations equal” and “only one generation exists” (Finding F78).

3.2 Predictivity

$Q = \frac{2}{3}$ plus two masses overdetermines the third:

$$m_\tau^{\text{pred}} = 1776.97 \text{ MeV} \quad \text{vs} \quad m_\tau^{\text{PDG}} = 1776.86 \pm 0.12 \text{ MeV} \quad (6.1 \times 10^{-5}).$$

(We re-verified $Q = 0.6666605$ from the PDG masses $m_e = 0.51099895$, $m_\mu = 105.6583755$, $m_\tau = 1776.86 \text{ MeV}$ for this paper.)

3.3 Why $\sqrt{m}$: the Cooper-pair bilinear

The amplitude coordinate is $\sqrt{m}$, not m , because the model treats the symmetry-breaking sector as superconductor-like: the would-be Higgs is a Cooper pair (Papers II-III). Applying the same premise to fermion mass, generation a ’s mass is a pair condensate, a bilinear in the constituent amplitude y_a :

$$m_a \propto \langle y_a y_a \rangle = y_a^2 \implies \sqrt{m_a} = y_a = \text{the } T_{1u} \text{ vector component on axis } a.$$

Koide is then a statement about the amplitude y , not the mass, precisely because mass is quadratic in the pairing amplitude (Finding F78, Part A — a derivation, not a fit). The data independently single out the bilinear power: the participation ratio $(\sum m^s)^2 / \sum m^{2s}$ is the clean rational $3/2$ only at $s = \frac{1}{2}$ (measured 1.500014; neighbours 1.834 at $s = \frac{1}{3}$, 1.119 at $s = 1$).

3.4 The 45° is the bound-pair cap — one $SO(2)$ rotation

The 45° of Koide is the *same* equipartition angle that caps a two-constituent bound state (Finding F73, used in Paper XI). Both of the model’s mass “rotations” are a unit 2-vector turned by an angle, whose two squared weights ($\cos^2, \sin^2$) become equal at 45°:

- **Constituent (bound-pair) level:** the constituent carries $(n_c, m_c) = (\cos t, \sin t)$ (kinetic complement and rest mass). The pair phase $\Omega_{\text{pair}} = 2t$ saturates the stability bound $\Omega = \pi/2$ exactly at $t = 45^\circ$, i.e. $m_c = 1/\sqrt{2}$, where rest and kinetic weights equipartition.
- **Generation level:** the amplitude carries (common, diff) = $(\cos \phi, \sin \phi)$ (A_{1g} vs T_{1u}); Koide equipartition is $\phi = 45^\circ$.

Numerically the two states are the identical 45°-rotated unit vector $(1/\sqrt{2}, 1/\sqrt{2})$, with the pair phase exactly $\pi/2$ there (Finding F80, D2). The constituent-level 45° and the generation-level 45° are one and the same $SO(2)$ equipartition, appearing once per level. Finding F81 sharpens the value: 45° is the two-constituent pair’s half of the $\pi/2$ phase budget — the spin-0 Cooper singlet $(\uparrow\downarrow - \downarrow\uparrow)/\sqrt{2}$ is itself a maximally-mixed 45° state, and the generation equipartition is plausibly the image of that internal structure.

4. The neutrino sector: a Higgs-free see-saw

4.1 The Majorana mass step

A singlet ν_R admits a bare Majorana mass $\mathcal{L}_M = -\frac{1}{2}M_R(\nu_R^T \varepsilon \nu_R + \text{h.c.})$, $\varepsilon = i\sigma^2$. In QCA form the Majorana step on one Weyl singlet is the closed-form integration of the BdG equations of motion $i\partial_t \chi_u = +M_R \chi_d^*$, $i\partial_t \chi_d = -M_R \chi_u^*$:

$$\boxed{\chi'_u = c_M \chi_u - i s_M \chi_d^*, \quad \chi'_d = c_M \chi_d + i s_M \chi_u^*,} \quad c_M = \cos(M_R dt), \quad s_M = \sin(M_R dt).$$

The step is **anti-linear** (couples χ to χ^*) — intrinsic to a Majorana mass and the source of lepton-number violation. It is exactly R-unitary on the four real degrees of freedom ($|\chi'_u|^2 + |\chi'_d|^2 = |\chi_u|^2 + |\chi_d|^2$; verified over $M_R \in [0, 10^6]$, $dt \in [0.05, 1.27]$, 200-step compose, residual 2.3×10^{-11}). The $U(1)_Y$ selection rule is exact: the bilinear $\chi^T \varepsilon \chi$ picks up $e^{i\beta Y_{\nu_R}}$, so the step is gauge-invariant **iff** $Y_{\nu_R} = 0$ — the *same* hypercharge constraint that forbids a Higgs scalar, now forcing ν_R to be a $Y = 0$ singlet for a Majorana mass to exist (Finding F47). In the SM this is presented as a coincidence of the hypercharge assignment; here it is a consequence of one structural choice.

4.2 See-saw scaling

Combined with the chiral Dirac mass step, the (ν_L, ν_R^c) sector has the canonical see-saw matrix and eigenvalues

$$M = \begin{pmatrix} 0 & M_D \\ M_D & M_R \end{pmatrix}, \quad \lambda_{\pm} = \frac{M_R \pm \sqrt{M_R^2 + 4M_D^2}}{2}, \quad |\lambda_-| \xrightarrow{M_R \gg M_D} \frac{M_D^2}{M_R}.$$

The lattice (8\$×\$8 BdG) light eigenvalue matches the closed form to $\leq 10^{-12} M_R$, and the see-saw scaling $m_\nu \cdot M_R/M_D^2 \rightarrow 1$ holds within the next-order $(M_D/M_R)^2$ bound at every sampled ratio $M_R/M_D \in \{3, 10, 30, \dots, 10^5\}$ (Finding F47, M1-M6, 6/6). Concretely, with $M_D \sim m_e$ and $M_R \sim 10^{12}\text{--}10^{15}$ GeV, the see-saw produces $m_\nu \sim 10^{-2}\text{--}10^{-5}$ eV — the observed range — automatically. **The smallness of the neutrino mass is explained Higgs-free, by a single large scale M_R rather than an unnaturally tiny Yukawa.**

4.3 Three generations and mixing (outlook)

The single-flavour block generalises to a 3×3 Dirac matrix M_D and a 3×3 Majorana matrix M_R ; the 6×6 see-saw then yields three light (active) and three heavy (sterile) neutrinos, with PMNS mixing from the diagonalisation. The large *leptonic* mixing (near tribimaximal PMNS), in contrast to the small quark CKM mixing, is the democratic signature of the neutral sector — consistent with the neutrinos sitting *off* the charged-lepton critical point (§5). A full 3×3 build is flagged as the natural next step.

5. Why only the charged leptons equipartition: the electromagnetic selection rule

The charged leptons sit exactly at the 45° critical point; the quarks and neutrinos do not. The model's answer (Finding F80) is that **electric charge selects the sector**: only the charged leptons are simultaneously EM-coupled *and* colour-free, so only they land on the clean abelian critical point.

sector	EM coupling	Q	at 2/3?
charged leptons e, μ, τ	full ($q = -1$), no colour	0.66666	yes (45°)
up-type quarks u, c, t	$q = +\frac{2}{3}$, + colour/QCD	0.849	no
down-type quarks d, s, b	$q = -\frac{1}{3}$, + colour/QCD	0.731	no

sector	EM coupling	Q	at 2/3?
neutrinos ν	none ($q = 0$; Majorana/see-saw)	ordering-dependent 0.34-0.59	no (unpinned)

Quarks carry the EM rotation too, but their mass dynamics are dominated by the non-abelian QCD condensate, which pushes them off the clean EM point — and the two quark charges give two *different* Q 's ($0.85 \neq 0.73$), so there is no universal value. Neutrinos have no EM rotation at all, so nothing pins them to $\frac{2}{3}$, and indeed their Koide ratio is not yet well-defined (it swings with the unknown lightest mass and ordering).

The honest residual. Electromagnetism explains *which* sector sits at the critical point, but not the *magnitude* of the rotation. A perturbative EM self-energy supplies only $\Delta\phi_{\text{EM}} \sim \alpha/\pi = 0.13^\circ$, whereas the rotation from democracy to equipartition is a full 45° — a factor ~ 340 too weak. The value 45° is therefore a *non-perturbative critical/saturation condition*, pinned by the data and by the recurring $1/\sqrt{2}$ of the Cooper-pair singlet, but not yet derived from a non-perturbative EM-driven gap. This is the single sharpest open problem of the lepton sector, and we record it rather than overclaim.

6. Verification summary

Result	Section	Residual / status
$T_3 = +\frac{1}{2}$ for pure ν_L (Higgs-free Y)	§2	1.1×10^{-16}
$U(1)_Y$ Ward identity (Stueckelberg kinetic)	§2	1.8×10^{-15}
Koide map $Q(\phi) = 1/(3 \cos^2 \phi)$; $45^\circ \rightarrow \frac{2}{3}$	§3.1	exact at $0/45/54.74^\circ$
Charged-lepton Q	§3.1	0.6666605 (0.91σ)
Bilinear power $s = \frac{1}{2}$ uniquely clean	§3.3	$1/Q = 1.500014$
$SO(2)$ unification of F73 cap & Koide 45°	§3.4	identical ($1/\sqrt{2}, 1/\sqrt{2}$); $\Omega_{\text{pair}} = \pi/2$
m_τ prediction	§3.2	6.1×10^{-5}
Majorana step	§4.1	2.3×10^{-11}
R-unitarity		
$U(1)_Y$ selection $Y_{\nu_R} = 0$	§4.1	0 exact at $Y = 0$
See-saw scaling $m_\nu \rightarrow M_D^2/M_R$	§4.2	within $(M_D/M_R)^2$ bound, all ratios

Result	Section	Residual / status
BdG light eigenvalue vs closed form	\$4.2	$\leq 10^{-12} M_R$
EM selection contrast (leptons/quarks/ ν)	\$5	0.667 vs 0.85, 0.73, unpinned

Underlying findings: F41 (Higgs-free hypercharge), F42 (dynamical χ kinetic), F47 (Majorana see-saw), F78 (Koide amplitude / Cooper bilinear), F80 (45° / EM selection), F81 (45° as half the phase budget), F75/F76 (three generations / hierarchy).

7. Discussion

The lepton sector ties together three of the model’s signatures. Hypercharge is Higgs-free because it rides the same gauge field as the mass phase. The Koide relation — the most precise unexplained lepton mass relation — becomes an exact statement: $Q = \frac{2}{3}$ is the 45° equipartition of the cubic amplitude $\sqrt{m}$, the same $SO(2)$ equal-split that caps the bound pair, and electromagnetism is the rule that selects the colour-free charged sector to sit there. The neutrino mass is small for the see-saw reason, realised Higgs-free with the $Y_{\nu_R} = 0$ constraint structural rather than coincidental.

Two open problems are stated sharply: (i) the *magnitude* of the 45° Koide rotation is a non-perturbative critical condition not yet derived (perturbative EM is $\sim 340\times$ too weak); (ii) the full 3×3 Dirac+Majorana neutrino build, with PMNS mixing, is not yet done. Neither affects the verified structure of the sector.

References

1. Y. Koide, “New view of quark and lepton mass hierarchy,” *Phys. Rev. D* **28**, 252 (1983); *Lett. Nuovo Cimento* **34**, 201 (1982).
2. P. Minkowski, *Phys. Lett. B* **67**, 421 (1977); M. Gell-Mann, P. Ramond, R. Slansky (1979); T. Yanagida (1979); R. N. Mohapatra, G. Senjanović, *Phys. Rev. Lett.* **44**, 912 (1980) — see-saw mechanism.
3. Y. Nambu, “Quasi-particles and gauge invariance in the theory of superconductivity,” *Phys. Rev.* **117**, 648 (1960) — pairing / BdG.
4. Particle Data Group, *Review of Particle Physics* (2024) — charged-lepton and neutrino mass data; PMNS.
5. Project findings: F41, F42, F47, F73, F75, F76, F78, F80, F81.

Companion papers: III (mass / Koide), VI (weak isospin / Majorana extension), IX (fermion structure / generations), XI (baryons / bound-pair cap).